AMDARI

Cloud Security Internship Programme

Implementing Cloud Threat Detection and Automated Response with Microsoft Sentinel

Microsoft Sentinel SIEM/SOAR deployment for CloudScale Logistics

Prepared by Sunday Abiodun

AMDARI | CloudScale Logistics

Report

Implementing Cloud Threat Detection and Automated Response with Microsoft Sentinel

Implementation Report | CloudScale Logistics | Microsoft Azure, East US

Field	Detail
Analyst	Sunday Abiodun
Client	CloudScale Logistics (mid-market freight and supply chain)
Engagement	Microsoft Sentinel SIEM and SOAR, two-week proof of concept
Primary region	East US
Frameworks	NIST Cybersecurity Framework, MITRE ATT&CK, SOC 2 Type II, ISO 27001

1.0 Executive Summary

CloudScale Logistics has significantly strengthened its cloud security capabilities by implementing a centralised Security Operations platform across its Microsoft Azure environment. Previously, security monitoring was fragmented across multiple Azure services, limiting visibility, increasing investigation times, and making it difficult to respond effectively to potential security incidents.

To address these challenges, Microsoft Sentinel was deployed as the organisation's Security Information and Event Management (SIEM) and Security Orchestration, Automation and Response (SOAR) solution. Azure security logs were consolidated into a single Log Analytics Workspace, analytics rules were configured to detect suspicious activity based on established attack techniques, and an automated response playbook was implemented to ensure high-priority incidents are escalated to the security team within minutes.

The solution provides continuous monitoring of the Azure estate, enabling faster threat detection, streamlined investigations, and improved operational efficiency. It also establishes a scalable foundation for future security enhancements while supporting compliance with organisational governance requirements.

Business Value

Environment Before Deployment

- Security monitoring was decentralised, with no unified view of Azure security events.
- Suspected credential compromise incidents took more than four hours to progress from initial alert to containment decisions.
- Administrative and operational activities were distributed across multiple Azure portals, resulting in limited audit visibility and slower investigations.

Environment After Deployment

- All Azure control-plane activities are centrally collected, retained, and searchable within a single Log Analytics Workspace.
- Security incidents are automatically detected using Microsoft Sentinel analytics rules aligned to recognised attacker behaviours.
- High-priority incidents trigger automated notifications, enabling the security team to begin investigation within two minutes of incident creation.

- Security operations have evolved from a reactive monitoring approach to a proactive, intelligence-driven detection and response capability.

The implementation directly addresses the ISO 27001 audit finding that initiated the engagement while providing CloudScale Logistics with a robust, cloud-native security monitoring capability that supports ongoing operational resilience.

Key Benefits

- Centralised visibility across the entire Azure environment.
- Automated threat detection and incident response.
- Faster incident identification and reduced response times.
- Enhanced audit readiness through consolidated logging and monitoring.
- Scalable security platform supporting future cloud expansion and security maturity.

2.0 Project Context

Background

CloudScale Logistics migrated its core business platforms to Microsoft Azure during the eighteen months preceding this engagement. While migration successfully improved scalability and operational flexibility, it also expanded the organisation's attack surface across multiple Azure subscriptions. At the same time, the internal IT team remained primarily experienced in managing traditional on-premises infrastructure, creating challenges in securing and monitoring the new cloud environment.

Challenges Identified

Limited Security Visibility

Security data, including Azure Activity Logs, Microsoft Defender alerts, and virtual machine events—was distributed across multiple services with no centralised monitoring capability. The lack of consolidated visibility made it difficult to identify suspicious activity across subscriptions, increasing the risk that attackers could move laterally without being detected promptly.

Manual and Inefficient Incident Response

The security team relied on manual processes to review and investigate hundreds of alerts each day. Without automated correlation or response capabilities, analysts had to assess alerts individually, while incident escalation was managed through email. During one confirmed security incident, it took more than four hours to reach a containment decision, highlighting significant delays in the response process.

Compliance Gap

An ISO 27001 audit identified a non-conformance due to the absence of a documented cloud detection and incident response capability. The organisation was unable to demonstrate a consistent, auditable process for monitoring, detecting, and responding to security events within its Azure environment.

Proof of Concept Objective

The proof of concept (PoC) was designed to validate that a centralised and automated security monitoring and incident response capability could be rapidly deployed within Microsoft Azure. A secondary objective was to produce the operational evidence and documentation required to address the ISO 27001 audit finding and support compliance requirements.

3.0 Solution Architecture

The proposed solution is built around three core functions aligned with the Detect, Respond, and Recover phases of the NIST Cybersecurity Framework. The architecture centralises security telemetry, correlates threats across the Azure environment, and automates incident response through a streamlined, one-way data flow. This approach simplifies security operations while providing a transparent and auditable detection and response process.

> **Data flow (one line)**
> Azure data sources -> Log Analytics Workspace -> Sentinel analytics rules -> Incident -> Logic App playbook -> Team notification

In plain terms: Azure Activity logs and Microsoft Defender for Cloud feed raw security telemetry into the Log Analytics Workspace named LAW-Sentinel-CloudScale. Microsoft Sentinel sits on top of that workspace and runs analytics rules written in Kusto Query Language (KQL) against the incoming data. When a rule matches, Sentinel raises an incident. The incident triggers a Logic App playbook, which formats the details and posts them to the security team's notification channel. Every stage leaves a record in the workspace, which is what gives the auditor a full trail.

3.1 Component roles

Component	Role
Azure Resource Group (RG-Sentinel-PoC)	Provides a logical container for all project resources, enabling organised resource management, access control, and lifecycle management.
Log Analytics Workspace (LAW-Sentinel-CloudScale)	Acts as the centralised data repository for security telemetry, storing logs collected from Azure services and integrated security solutions.
Microsoft Sentinel	Provides SIEM and SOAR functionality, including security monitoring, analytics, incident management, threat investigation, and automation.
Azure Activity Logs Connector	Collects Azure subscription-level activity data, including administrative operations, resource changes, and role assignments.
Microsoft Defender for Cloud Connector	Provides high-quality security alerts and threat intelligence from Azure workloads and services.
Analytics Rules	Analyse security events and generate alerts based on known threats, suspicious behaviours, and custom detection logic.
KQL Detection Rules	Provide customised threat detection by querying security telemetry for specific attack scenarios such as privilege escalation.
Microsoft Sentinel Incidents	Aggregate related alerts into actionable security cases for investigation and response.
Azure Logic App Playbooks	Automate response actions such as sending notifications and initiating security workflows when incidents occur.
Microsoft Teams / Email Integration	Provides communication channels for security teams to receive real-time incident notifications.

4.0 Implementation

The implementation phase delivers the Microsoft Sentinel security monitoring solution through a structured deployment approach. The process was divided into three key stages: Foundation, Data Ingestion, and Detection Engineering. Each stage builds upon the previous phase to establish a complete security monitoring and response capability.

4.1 Foundation

The foundation phase establishes the core Azure infrastructure required for Microsoft Sentinel deployment and operation.

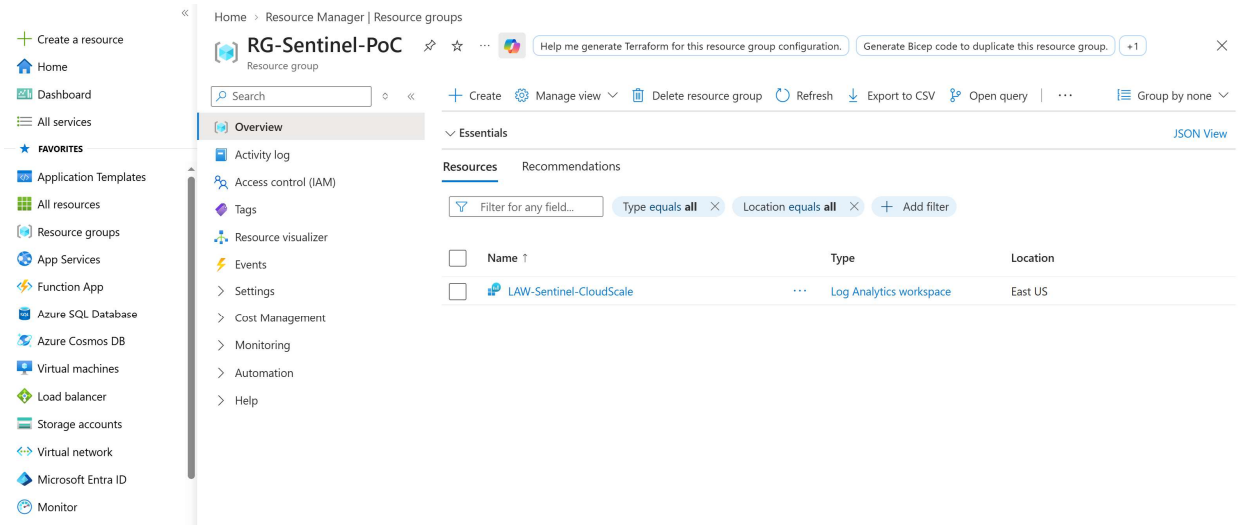
The implementation begins by creating the RG-Sentinel-PoC Resource Group within the East US region. This resource group acts as a central management boundary for all Sentinel-related resources, allowing simplified administration, access control, and lifecycle management.

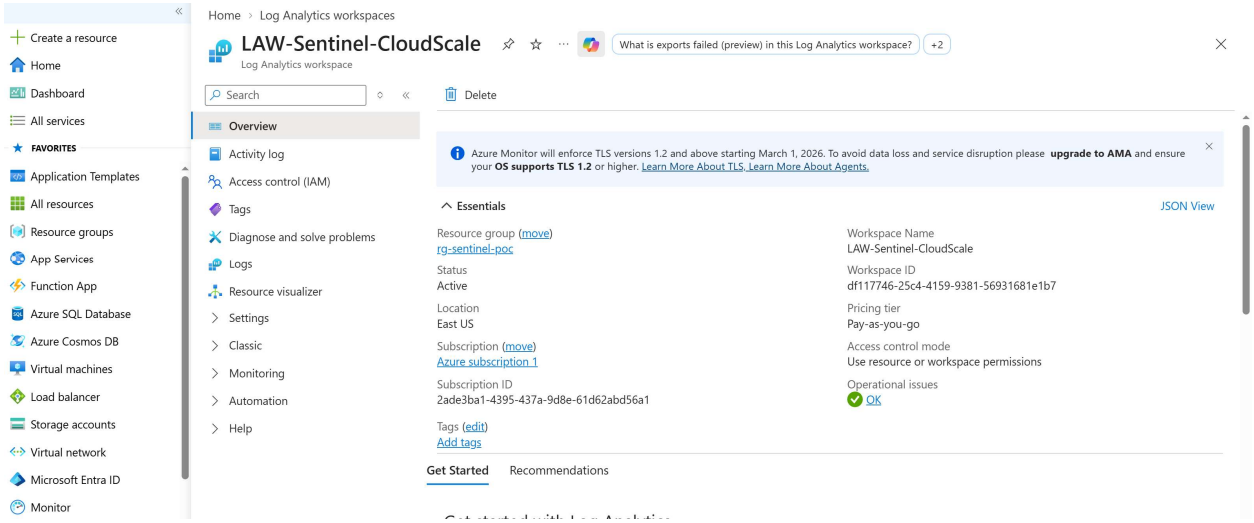
A Log Analytics Workspace named LAW-Sentinel-CloudScale is deployed within the resource group. This workspace provides a centralised data storage platform for collecting, storing, and analysing security telemetry. The workspace is configured with appropriate retention settings to support security investigation and compliance requirements.

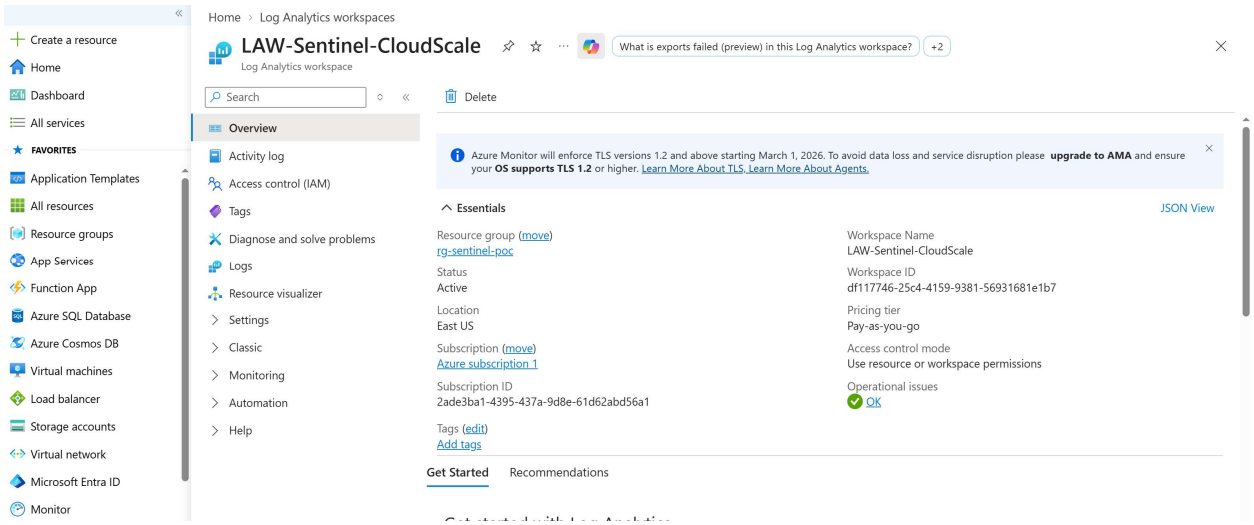
was then enabled on the Log Analytics Workspace, activating SIEM and SOAR capabilities. This provides the environment with security analytics, incident management, threat hunting, and automated response functionality.

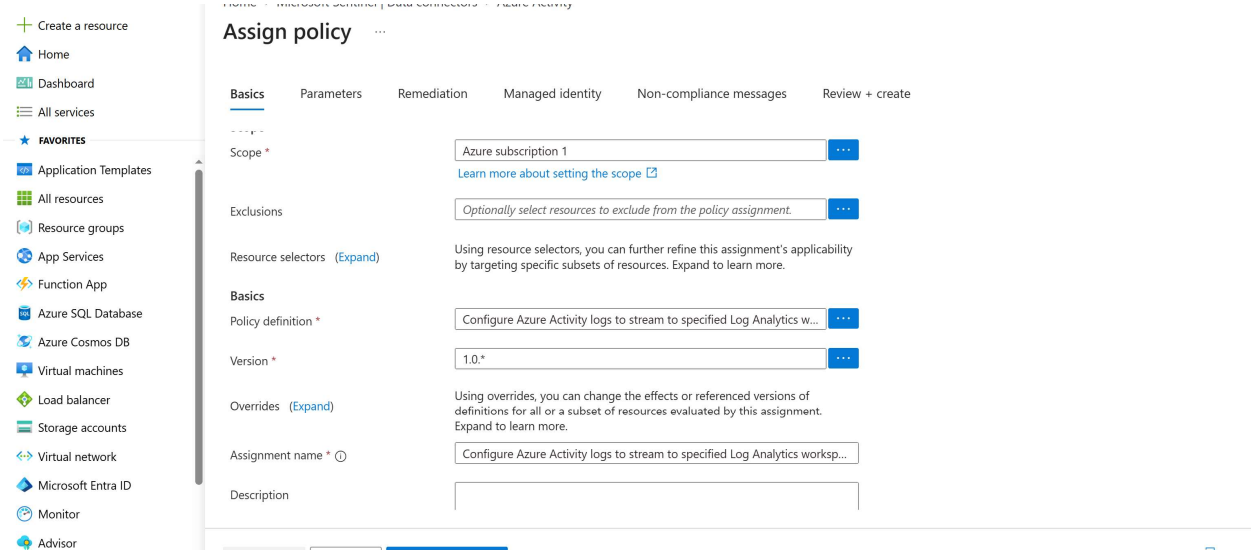
The foundation phase is validated by confirming:

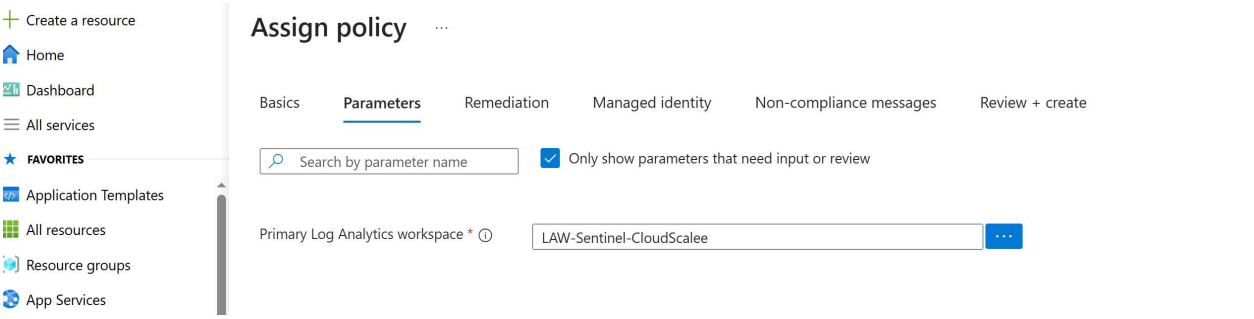
- Resource Group deployment is successful
- Log Analytics Workspace is active
- Microsoft Sentinel is enabled and accessible
- Workspace configuration is ready for data connection


Home › Resource Manager | Resource groups
RG-Sentinel-PoC
Resource group
Help me generate Terraform for this resource group configuration.
Generate Bicep code to duplicate this resource group.
+1
Create
Manage view
Delete resource group
Refresh
Export to CSV
Open query
Group by none
Essentials
JSON View
Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Cost Management
Monitoring
Automation
Help
Resources
Recommendations
Filter for any field...
Type equals all
Location equals all
Add filter
Name
Type
Location
LAW-Sentinel-CloudScale
Log Analytics workspace
East US
Create a resource
Home
Dashboard
All services
FAVORITES
Application Templates
All resources
Resource groups
App Services
Function App
Azure SQL Database
Azure Cosmos DB
Virtual machines
Load balancer
Storage accounts
Virtual network
Microsoft Entra ID
Monitor


Home › Log Analytics workspaces
LAW-Sentinel-CloudScale
Log Analytics workspace
What is exports failed (preview) in this Log Analytics workspace?
+2
Delete
Azure Monitor will enforce TLS versions 1.2 and above starting March 1, 2026. To avoid data loss and service disruption please upgrade to AMA and ensure your OS supports TLS 1.2 or higher. Learn More About TLS, Learn More About Agents.
Essentials
JSON View
Resource group (move)
rg-sentinel-poc
Status
Active
Location
East US
Subscription (move)
Azure subscription 1
Subscription ID
2ade3ba1-4395-437a-9d8e-61d62abd56a1
Tags (edit)
Add tags
Workspace Name
LAW-Sentinel-CloudScale
Workspace ID
df117746-25c4-4159-9381-56931681e1b7
Pricing tier
Pay-as-you-go
Access control mode
Use resource or workspace permissions
Operational issues
OK
Get Started
Recommendations
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Logs
Resource visualizer
Settings
Classic
Monitoring
Automation
Help


Home › Log Analytics workspaces
LAW-Sentinel-CloudScale
Log Analytics workspace
What is exports failed (preview) in this Log Analytics workspace?
+2
Delete
Azure Monitor will enforce TLS versions 1.2 and above starting March 1, 2026. To avoid data loss and service disruption please upgrade to AMA and ensure your OS supports TLS 1.2 or higher. Learn More About TLS, Learn More About Agents.
Essentials
JSON View
Resource group (move)
rg-sentinel-poc
Status
Active
Location
East US
Subscription (move)
Azure subscription 1
Subscription ID
2ade3ba1-4395-437a-9d8e-61d62abd56a1
Tags (edit)
Add tags
Workspace Name
LAW-Sentinel-CloudScale
Workspace ID
df117746-25c4-4159-9381-56931681e1b7
Pricing tier
Pay-as-you-go
Access control mode
Use resource or workspace permissions
Operational issues
OK
Get Started
Recommendations
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Logs
Resource visualizer
Settings
Classic
Monitoring
Automation
Help



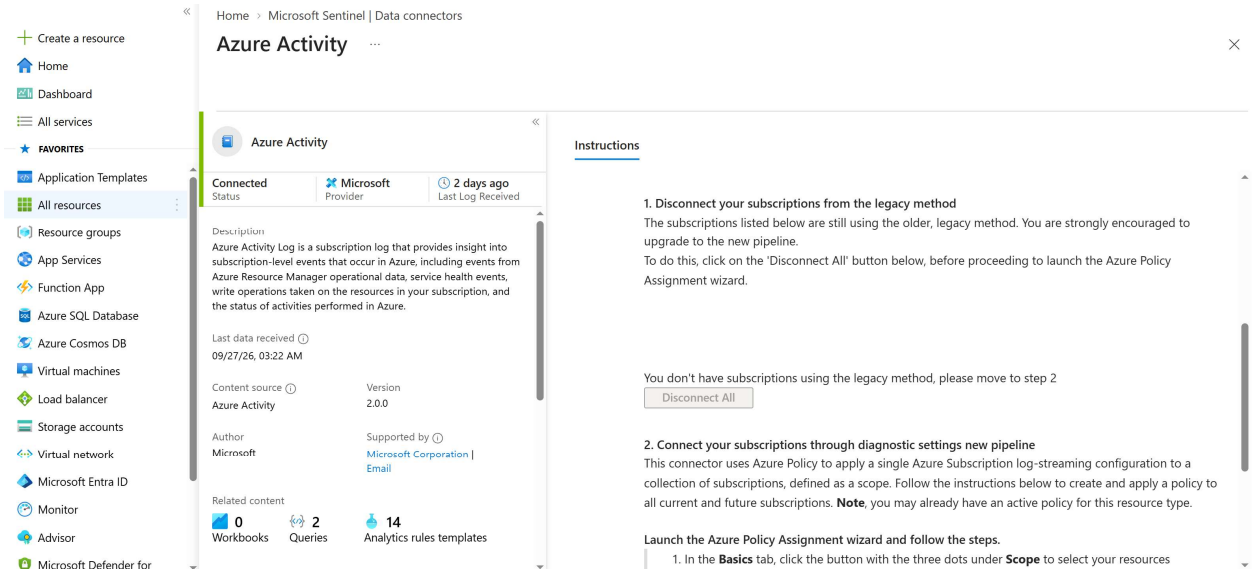


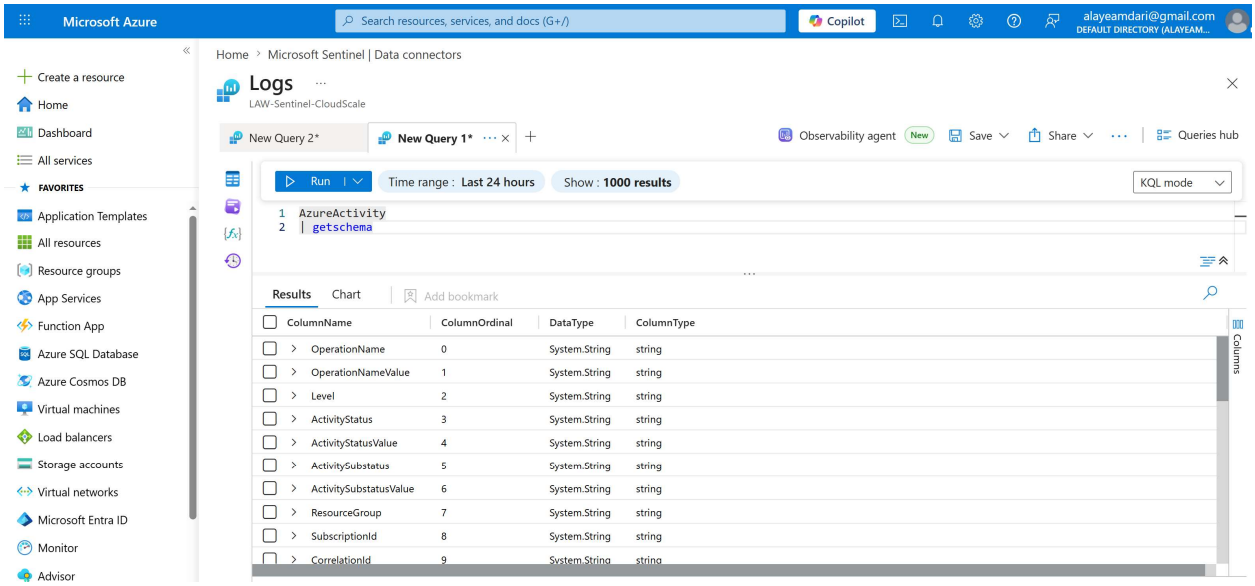
4.2Data ingestion

I enabled two data connectors for proof of concept. The **Azure Activity** connector collects all control-plane activity across the subscription, providing the telemetry required for our custom detection rules. The **Microsoft Defender for Cloud** connector imports pre-analyzed security alerts generated from monitored workloads.

To keep the proof of concept cost-effective, we used the free foundational tier, which allowed us to evaluate the solution with minimal expense. For a production deployment, however, the paid Defender plans are recommended to provide broader security coverage and advanced threat detection capabilities.

Before creating any detection rules, we verified that data ingestion was working correctly. This validation step is critical because detection rules cannot generate alerts if the underlying data tables contain no records. Approximately 30 minutes after enabling the connectors, both test queries returned live data, confirming that the data pipeline was functioning as expected.






```
// Confirm control-plane data is arriving
AzureActivity
| take 10

// Confirm Defender alerts are arriving
SecurityAlertake 10
```

4.3 Detection engineering

I configured ten analytics rules: nine built-in scheduled rules that monitor common attack techniques, along with one custom rule designed specifically for the client's environment. The built-in rules detect suspicious activities such as brute-force login attempts, connections from known malicious IP addresses, and unusual resource deletion events. Each rule is mapped to the relevant MITRE ATT&CK tactic, providing valuable context about the attacker's objectives rather than just generating raw alerts.

The custom rule addresses the client's primary security concern: unauthorized privilege escalation by an insider or a compromised account. It generates an alert whenever an Owner or Contributor role is assigned outside normal business hours, helping to identify suspicious privilege assignments that may indicate malicious activity. The first version looked like this

```
AzureActivity
| where OperationNameValue == "MICROSOFT.AUTHORIZATION/ROLEASSIGNMENTS/WRITE"
| where ActivityStatusValue == "Success"
| extend Role = tostring(parse_json(Authorization).evidence)
| where Role contains "Owner" or Role contains "Contributor"
| extend Hour = hourofday(TimeGenerated)
| extend DayOfWeek = dayofweek(TimeGenerated)
| where Hour < 8 or Hour > 18 or DayOfWeek == 0d or DayOfWeek == 6d
| project TimeGenerated, Caller, Role, ResourceGroup, SubscriptionId
```

:

4.4 Incident grouping

Without incident grouping, one security event may trigger multiple separate alerts, leading to a high volume of notifications and making it difficult for analysts to distinguish real security incidents from normal activity. To enhance incident handling and minimise alert fatigue, incident grouping was enabled in Microsoft Sentinel. This feature combines related alerts into a single incident when they share similar characteristics, such as the same user account or IP address, within a specified 24-hour period.

By bringing connected alerts together into one consolidated incident, analysts gain a broader understanding of the potential threat activity instead of reviewing multiple disconnected alerts. This improves the efficiency of the investigation process by helping security teams assess the full attack

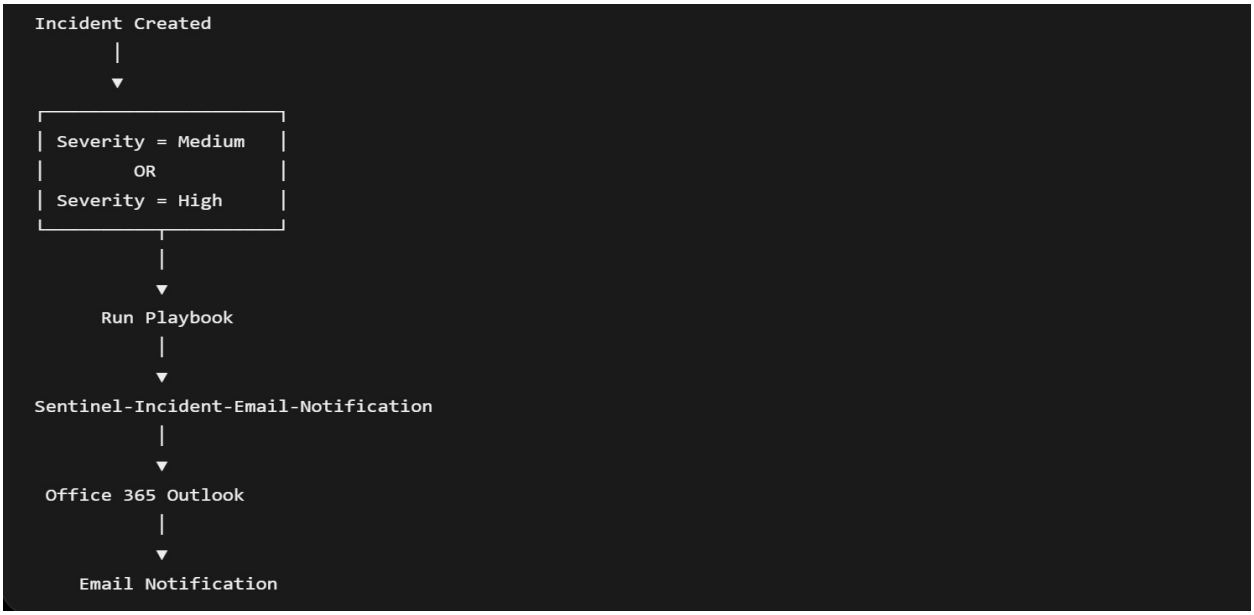
sequence, recognise threat patterns, and prioritise appropriate response measures. Consequently, incident analysis becomes faster, reduces the amount of manual investigation required, and supports more accurate threat detection and response.

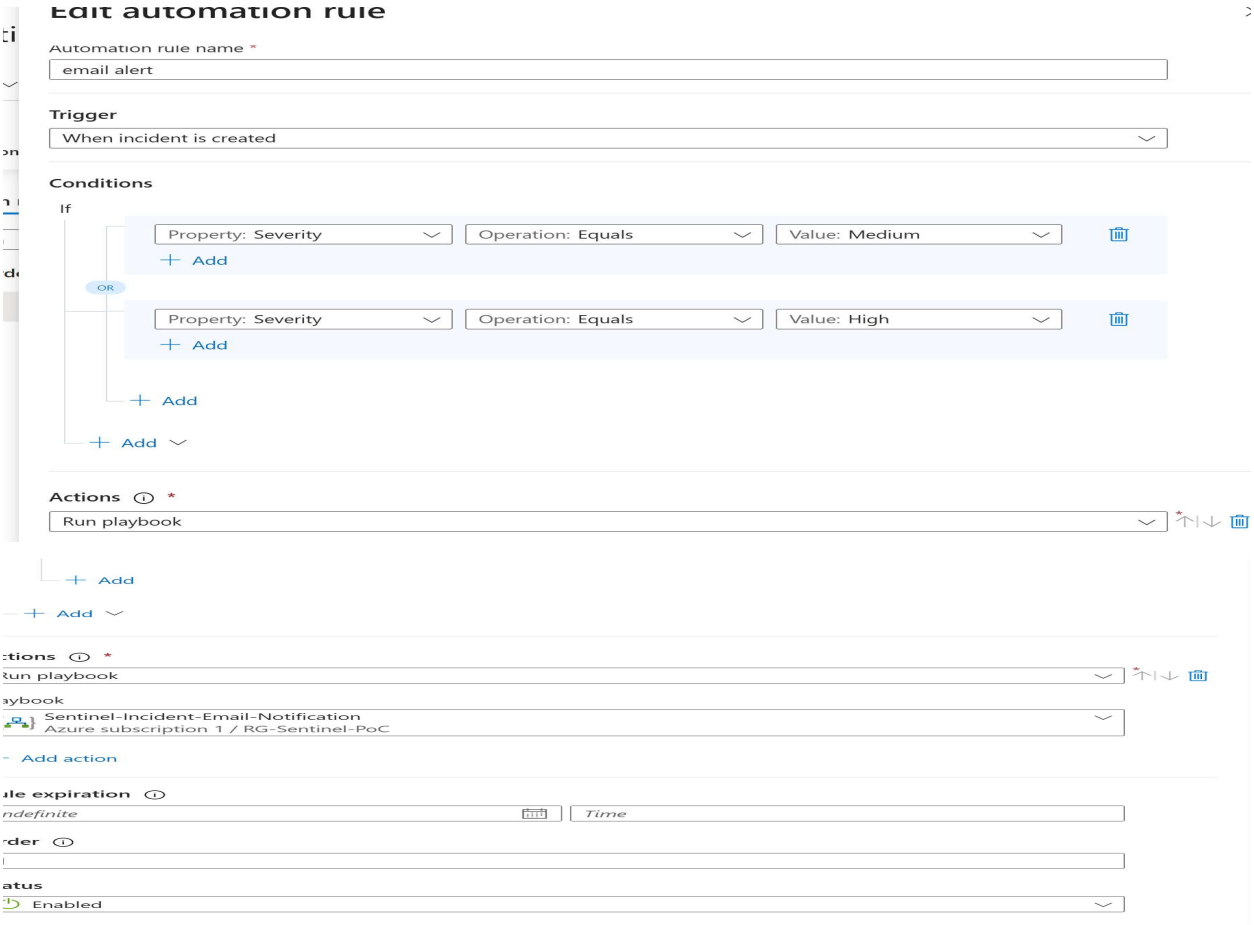
4.5 Automation

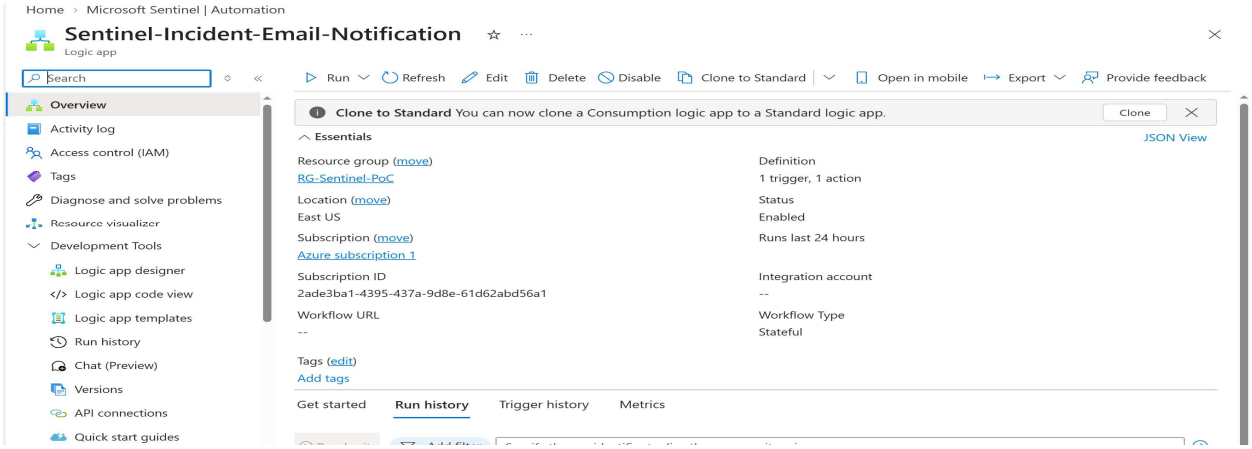
The final stage of the solution implementation integrates automated incident response capabilities with the detection framework. An Azure Logic App playbook was configured to automatically execute whenever Microsoft Sentinel generates a new incident. The playbook delivers a structured incident notification to the designated security team communication channel, ensuring that critical security events are communicated promptly without depending on continuous manual monitoring.

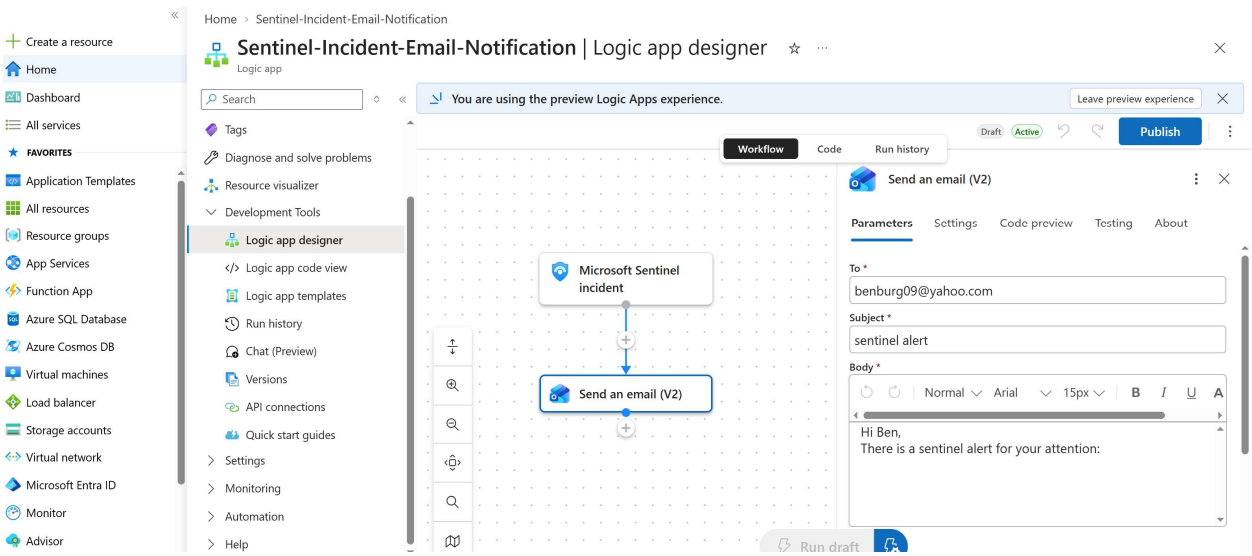
The automated notification includes essential incident information, such as the incident title, severity, affected entities, and relevant investigation details. This enables analysts to begin the response process immediately without first accessing the Microsoft Sentinel portal.

The Logic App playbook was subsequently connected to the custom analytics rule through the automated response configuration. This establishes an end-to-end security workflow where threat detection automatically initiates notification and response actions. Since the playbook operates independently from the individual analytics rule, it can be reused across additional detection scenarios, improving the scalability, maintainability, and operational efficiency of the security monitoring solution.


Incident Created
Severity = Medium
OR
Severity = High
Run Playbook
Sentinel-Incident-Email-Notification
Office 365 Outlook
Email Notification


Edit automation rule
Automation rule name *
email alert
Trigger
When incident is created
Conditions
If
Property: Severity
Operation: Equals
Value: Medium
Add
OR
Property: Severity
Operation: Equals
Value: High
Add
Add
Add
Actions *
Run playbook
Add
Add
Sentinel-Incident-Email-Notification
Azure subscription 1 / RG-Sentinel-PoC
Add action
Time
Enabled



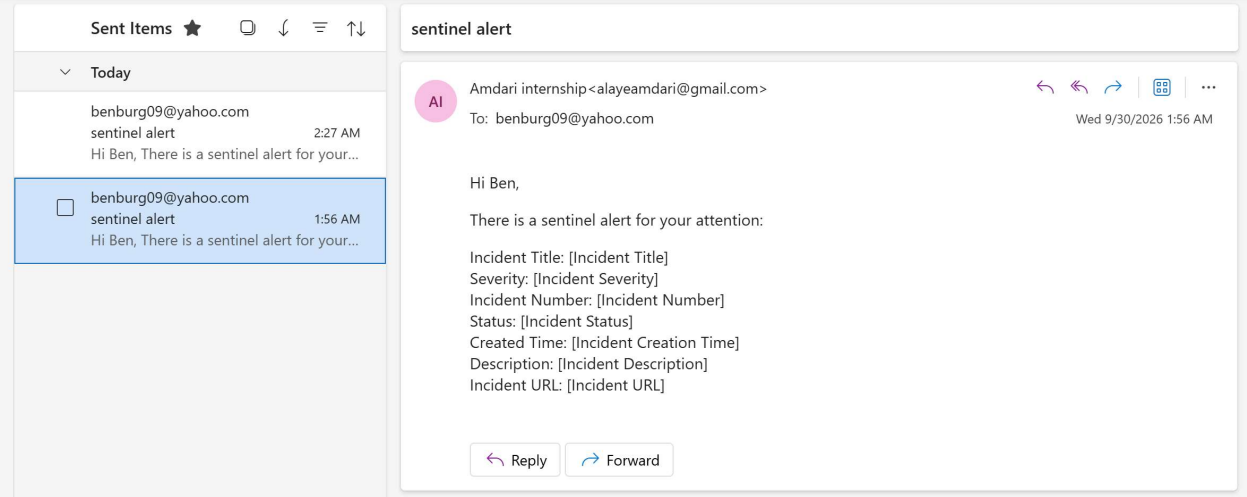



```
Incident: [IncidentTitle]
Severity: [Severity]
Time:     [TimeGenerated]
Entities: [Entities]
Link:     [IncidentUrl]   // deep link straight to the Sentnt
```

5. Testing and Validation

The system was successfully verified through an end-to-end controlled test. The test reproduced the specific threat scenario the custom rule was designed to detect: the assignment of a high-privilege role to a test account outside normal business hours. To ensure the alert was triggered during the testing session, the rule's look-back period was temporarily modified.

The complete detection workflow executed successfully, from the initial role assignment through to the delivery of the notification, in less than five minutes. When compared with the pre-engagement baseline, where containment decisions took more than four hours, the results demonstrate that the project achieved its intended objective of significantly improving detection and response speed.



Stage	What we observed	Evidence
Role assigned	Owner role granted to the test account	AzureActivity row for the roleAssignments/write operation
Rule fired	Custom rule matched within one run cycle	Alert visible in Logs and in the rule run history
Incident created	A single incident raised, correctly grouped	Incidents blade shows the new incident
Playbook ran	Logic App executed without error	Logic App run history shows a successful run
Team notified	Formatted message delivered with a working deep link	Notification received in the channel

6.0 Against Success Metrics

The metrics below were defined at the start of the engagement. Showing the goal alongside the actual outcome allows reviewers to evaluate success briefly.

Success metric	Target	Outcome
Centralised visibility	100% of admin actions queryable within 15 minutes	**Met**
Analytics coverage	10+ rules enabled, all mapped to MITRE ATT&CK	**Met**
Custom detection	Privilege escalation rule fires on simulation	**Met (after GUID fix)**
Automated notification	Playbook fires within 2 minutes of incident	**Met**
Audit retention	Telemetry retained 90 days minimum	**Met**

7.0 Cost

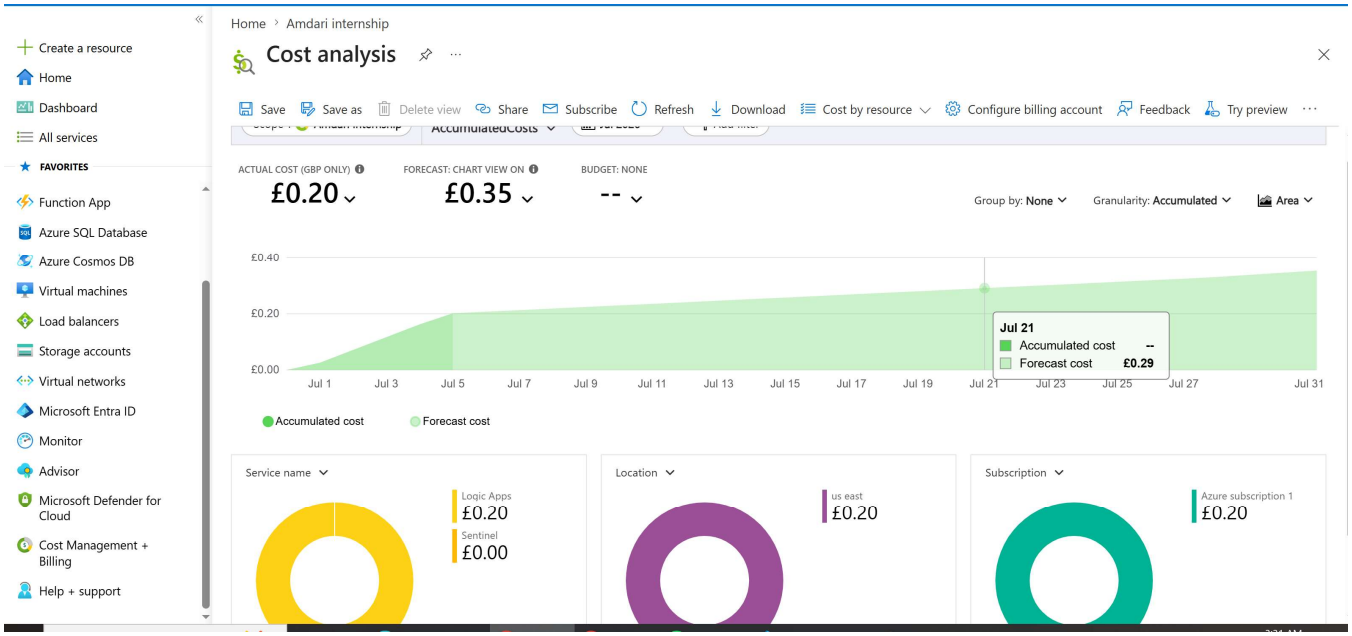
Sentinel's costs depend on usage:

Data ingestion per gigabyte

Data retention per gigabyte per month after the free period

Logic Apps execution per action

For this proof of concept, the cost was almost zero. Azure Activity logs ingest for free, the base Defender for Cloud tier is free, and the playbook ran only a few times, so Logic Apps charges were negligible. The main reason costs stayed near zero was the choice to defer paid Defender plans — which will be the key expense to plan for in



8.0 Challenges and Lessons Learned

 Verify **data ingestion status:** Perform initial validation queries to confirm that required data is being ingested before executing rules or analytics logic. Rules configured against empty datasets may not produce expected results and can lead to undetected failures.

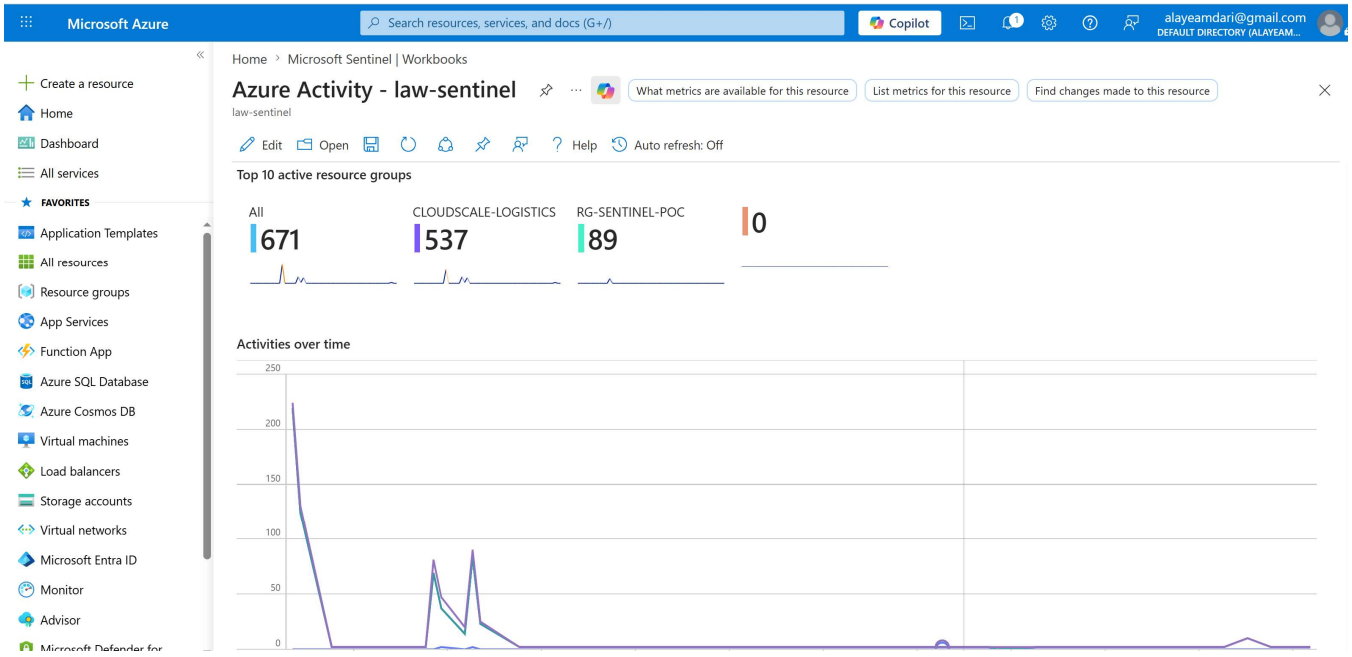
 Execute **end-to-end validation:** Component-level testing of individual rules or playbooks is insufficient to confirm operational readiness. Conduct full workflow testing to validate integration, dependencies, and execution flow across all components.

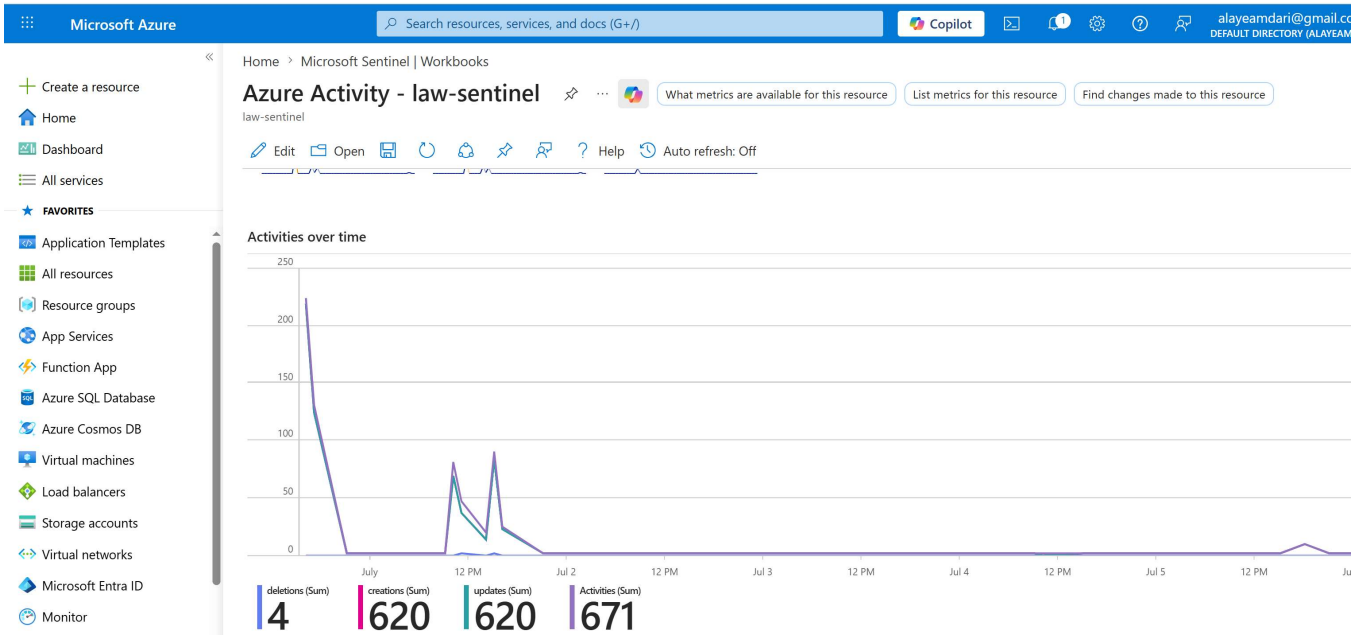
 Maintain **resource group organization:** Deploy related resources within a designated resource group to improve asset management, monitoring, access control, and lifecycle cleanup operations. Consistent resource organization reduces operational overhead.

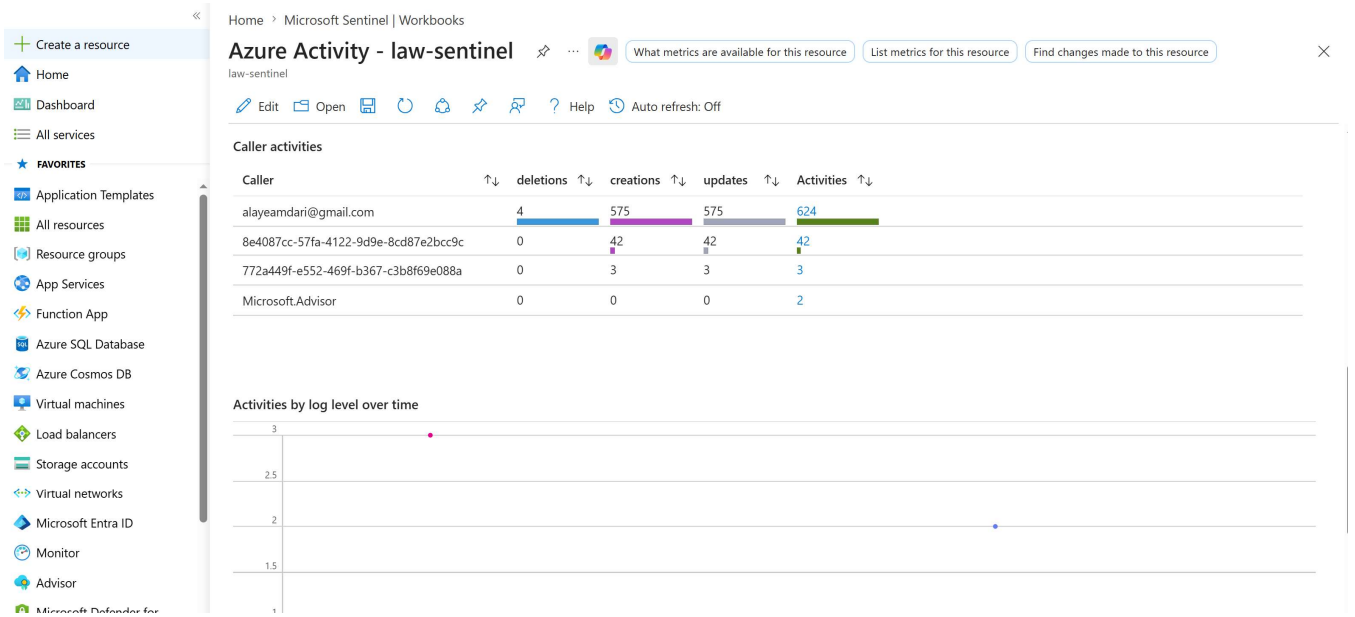
9.0 Future Enhancements

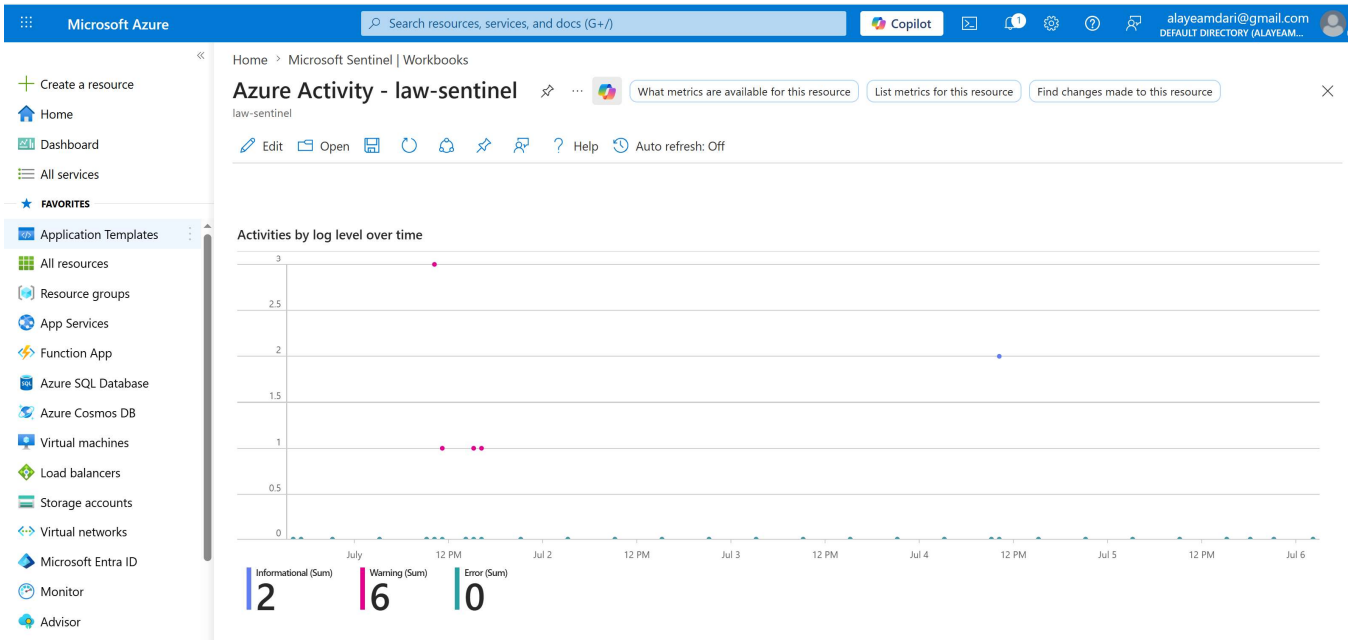
The solution is ready for production deployment and is designed to evolve through incremental improvements and added capabilities rather than major redesign. Planned enhancements include:

- Activate additional Microsoft Defender capabilities to improve protection across workloads and strengthen threat coverage.
- Introduce automated response and containment measures within the playbook, including approved actions such as disabling affected user accounts.
- Expand integration with additional security data sources, beginning with Microsoft Entra ID authentication logs and Microsoft 365 services.
- Create an executive-level security dashboard/workbook to provide stakeholders with a clear overview of security health and key metrics.
- Increase log retention capacity toward 180 days to enhance investigation, compliance, and audit readiness.




Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
alayeamdari@gmail.co
DEFAULT DIRECTORY (ALAYEAM
Create a resource
Home
Dashboard
All services
FAVORITES
Application Templates
All resources
Resource groups
App Services
Function App
Azure SQL Database
Azure Cosmos DB
Virtual machines
Load balancers
Storage accounts
Virtual networks
Microsoft Entra ID
Monitor
Home › Microsoft Sentinel | Workbooks
Azure Activity - law-sentinel
law-sentinel
What metrics are available for this resource
List metrics for this resource
Find changes made to this resource
Edit
Open
Help
Auto refresh: Off
Activities over time
250
200
150
100
50
0
July
12 PM
Jul 2
12 PM
Jul 3
12 PM
Jul 4
12 PM
Jul 5
12 PM
deletions (Sum)
4
creations (Sum)
620
updates (Sum)
620
Activities (Sum)
671


Create a resource
Home
Dashboard
All services
FAVORITES
Application Templates
All resources
Resource groups
App Services
Function App
Azure SQL Database
Azure Cosmos DB
Virtual machines
Load balancers
Storage accounts
Virtual networks
Microsoft Entra ID
Monitor
Advisor
Home › Microsoft Sentinel | Workbooks
Azure Activity - law-sentinel
law-sentinel
What metrics are available for this resource
List metrics for this resource
Find changes made to this resource
Edit
Open
Help
Auto refresh: Off
Caller activities
Caller
deletions
creations
updates
Activities
alayeamdari@gmail.com 4 575 575 624
8e4087cc-57fa-4122-9d9e-8cd87e2bcc9c 0 42 42 42
772a449f-e552-469f-b367-c3b8f69e088a 0 3 3 3
Microsoft.Advisor 0 0 0 2
Activities by log level over time
3
2.5
2
1.5



10.0 Conclusion

This project improved CloudScale Logistics' cloud security capabilities by moving from limited visibility to an automated approach for detecting and responding to critical threats within minutes. The three-pillar strategy — **collect, identify, and respond** — provided a clear framework for developing the solution and established a foundation for future security enhancements. The project also successfully delivered the documented detection and response process needed to resolve the ISO 27001 finding.

For security analysts, this project provided valuable hands-on experience in cloud security operations, including workspace configuration, KQL-based detection development, MITRE ATT&CK technique mapping, and SOAR automation. The key takeaway from creating the custom detection rule was that security is not achieved simply by implementing a solution; it requires continuous testing, validation, and clear documentation to ensure the process is reliable, understandable, and repeatable.